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Student Performance Analyzer in Java

Student Class

```
import java.util.*;

class Student {
    int id;
    String name;
    List<String> courses;
    Map<String, Integer> scores;

    public Student(int id, String name, List<String> courses,
Map<String, Integer> scores) {
        this.id = id;
        this.name = name;
        this.courses = courses;
        this.scores = scores;
    }

    public double getAverageScore() {
        return scores.values().stream()
.mapToInt(Integer::intValue)                .average()
        .orElse(0);
    }
}
```

Analyzer Class

```
import java.util.*;
import java.util.stream.*;

public class StudentAnalyzer {

    public static List<Student> getTopNStudents(List<Student>
students, int n) {
        return students.stream()
```

```

        .sorted(Comparator.comparingDouble(Student::getAverageScore).reversed())
            .limit(n)
            .collect(Collectors.toList());
    }

    public static Map<String, Double>
    getAverageScorePerCourse(List<Student> students) {

        Map<String, List<Integer>> courseScores = new
        HashMap<>();

        for (Student s : students) {
            for (String course : s.courses) {
                int score = s.scores.getDefault(course, 0);

                courseScores
                    .computeIfAbsent(course, k -> new
ArrayList<>())
                    .add(score);
            }
        }

        Map<String, Double> result = new HashMap<>();

        for (String course : courseScores.keySet()) {
            double avg = courseScores.get(course)
                .stream()
                .mapToInt(Integer::intValue)
                .average()
                .orElse(0);

            result.put(course, avg);
        }

        return result;
    }

    public static Set<String> getAllUniqueCourses(List<Student>
students) {

        return students.stream()
            .flatMap(s -> s.courses.stream())
            .collect(Collectors.toSet());
    }

```

```
    }  
}
```

Main Class

```
import java.util.*;  
  
public class Main {  
    public static void main(String[] args) {  
  
        List<Student> students = new ArrayList<>();  
  
        Map<String, Integer> s1Scores = new HashMap<>();  
        s1Scores.put("Math", 85);  
        s1Scores.put("Java", 90);  
        s1Scores.put("DSA", 80);  
  
        students.add(new Student(  
            1,  
            "Rahul",  
            Arrays.asList("Math", "Java", "DSA"),  
            s1Scores));  
  
        Map<String, Integer> s2Scores = new HashMap<>();  
        s2Scores.put("Math", 70);  
        s2Scores.put("Java", 75);  
        s2Scores.put("DSA", 65);  
  
        students.add(new Student(  
            2,  
            "Priya",  
            Arrays.asList("Math", "Java", "DSA"),  
            s2Scores));  
  
        Map<String, Integer> s3Scores = new HashMap<>();  
        s3Scores.put("Math", 95);  
        s3Scores.put("Java", 88);  
        s3Scores.put("DSA", 92);  
  
        students.add(new Student(  
            3,  
            "Amit",  
            Arrays.asList("Math", "Java", "DSA"),  
            s3Scores));  
  
        List<Student> topStudents =
```

```
StudentAnalyzer.getTopNStudents(students, 2);

    System.out.println("Top Students:");
    for (Student s : topStudents) {
        System.out.println(s.name + " Avg Score: " +
s.getAverageScore());
    }

    System.out.println("\nAverage Score Per Course:");

System.out.println(StudentAnalyzer.getAverageScorePerCourse(stu
dents));

    System.out.println("\nUnique Courses:");

System.out.println(StudentAnalyzer.getAllUniqueCourses(students
));
    }
}
```